

Caleb Theil • Full-Stack Engineer
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PROFESSIONAL SUMMARY

Full-stack engineer with deep experience in designing scalable, secure systems across the web, IoT, and infrastructure domains. Proven ability to apply computer science principles—such as distributed systems, concurrency, and systems architecture—to solve complex real-world problems. Self-motivated and academically driven, with a portfolio of impactful projects and production-grade systems.

EDUCATION

University of Colorado Denver Bachelor of Fine Arts in Digital Design (UX/UI Emphasis)
Graduated: May 2024

TECHNICAL SKILLS

Languages & Frameworks: TypeScript, JavaScript, Golang, SQL, HTML, CSS, React, Node.js, Express

Systems & Architecture: REST APIs, Distributed Systems, Caching, Concurrency

Databases: PostgreSQL, MongoDB, Redis, RDS, Valkey

Infrastructure & DevOps: AWS (EC2, RDS, EKS, S3, Lambda), Docker, GitHub Actions, Linux, NGINX, CI/CD, GCP, Terraform

Other Tools: Ansible, Rego, Styra, Jira, SAgile

EXPERIENCE

FULL-STACK ENGINEER | COBANK | HYBRID | JAN 2025 - PRESENT

Delivering technically rigorous solutions in distributed systems, scalable architecture, and performance engineering within a cross-functional Agile team.

- Led the **design and implementation** of a **scalable** user preferences **system** spanning multiple services, including a custom **write-through caching** layer, **trie-based lookup** logic, and **orchestration middleware** — **optimizing system performance and modularity**.
- Developed a **multi-tenant experience framework** by refactoring existing architecture, **enabling cross-user domain** control with **minimal disruption** to the core design.
- **Diagnosed and resolved production RDS bottlenecks and proxy misconfigurations**, collaborating with infrastructure teams to **restore stability** and **improve database performance**.
- **Introduced** team-wide **best practices for CI/CD pipelines**, structured logging, and **circuit breaker patterns**; **spearheaded major refactors** that improved maintainability and **reduced tech debt**.
- Championed adoption of secure, **server-side caching** and **policy-based access control** using **Redis and Rego (via Styra)**, **increasing system resilience and data integrity**.

FULL-STACK DEVELOPER - CONSULTANT | COLLEGE OF ARTS AND MEDIA | REMOTE | SEP 2024 - DECEMBER 2024

- **Diagnosed critical architectural bottlenecks** in **legacy systems** and proposed **scalable refactor strategies** aligned with **long-term maintainability**.
- **Implemented authentication and access control upgrades** to **safeguard sensitive user data** and **improve system integrity**.
- Built **custom data validation tools** to **automate error detection** and **improve reporting accuracy** across integrated systems.
- Partnered with stakeholders and infrastructure teams to **align application performance improvements with Agile workflows**.

FREELANCE FULL-STACK DEVELOPER | MAY 2024 - PRESENT | *Clients: Learning Resources Center, Guerilla Sports, College of Arts and Media, University of Northern Arizona*

- Built and deployed a digital signage platform (React, Node.js, Raspberry Pi) **replacing legacy infrastructure** with scalable, **remotely managed hardware**.
- **Developed a secure Golang API** and **automated provisioning using IaC** and a **ring-buffer-based logging solution** to ensure **long-term device reliability**.
- Developed a **versioned CDN delivery system** to **inject custom scripts** into a Webflow-based site, **enabling advanced features not supported by native capabilities**.

LEAD SOFTWARE DEVELOPER | LEARNING RESOURCES CENTER | DENVER, CO | APRIL 2021 - MAY 2024

- Led development of a React + **AWS Lambda admin platform** that improved load times by 80% and **increased staff efficiency**.
- Designed **scalable RESTful APIs** and **integrated RFID systems** for accurate, **data-driven operations**.

- Introduced Agile practices and **cross-team tooling** that **accelerated feature delivery** and **improved collaboration**.

PERSONAL PROJECTS

PMDB: Social app to help users discover movies/TV shows. Built with **Golang, Postgres, Redis, and React**, incorporating streaming service integration and real-time multi-user socket communication. Designed backend schema for recommendation scoring and session state synchronization

Echo: AI-driven interaction platform built with React and the **OpenAI API**. Deployed on **AWS EC2** with **NGINX** and SSL for secure, scalable functionality and low-latency API integrations.

HomeLab: Designed and deployed a custom-built server to orchestrate a home automation stack using **Proxmox** for virtualization, **TrueNAS** with RAIDZ for redundant storage, and **Home Assistant** for smart device management. Implemented secure ingress via a lightweight **Debian Bookworm** proxy routing

Cloudflare Tunnels into the internal network. Built container orchestration, logging, and automated backups to ensure uptime and fault recovery.